9 Cross-tabulating two qualitative variables

Sheet 49 Thu 11 Mar. 2027 *Cross-tabulating two qualitative variables*

1. Warm-up · Quick Express $\frac{24}{60}$ as a percentage.
2. Core · Quick In a class of 30 pupils, 18 are girls. What is the relative frequency of girls? of boys?
3. Core + · Classify Classify each variable: eye colour; baccalauréat grade (AB, B, TB); height in cm.
4. Review · Problem A rectangle of perimeter 20 cm has length x. Express its area $A(x)$; calculate $A(3)$ and $A(5)$.
5. Thread · Quick Calculate the exact volume of a cylinder of radius 2 cm and height 5 cm.

Sheet 50 Mon 15 Mar. 2027 *Qualitative variables; two-way frequency table*

1. Warm-up · Quick Complete: $12 + \ldots = 30$ and $22 + \ldots = 40$.
2. Core · Table Total 40 pupils. Girls: 22; canteen users (C): 25; girls who use the canteen: 15. Complete the two-way table.

	C	Not C	Total
Girls	15	…	22
Boys	…	…	…
Total	25	…	40

3. Core + · True/false True or false? Justify. "Satisfaction level (low, medium, high)" is a nominal qualitative variable.
4. Review · Problem Discount A: $-30\,\%$ on 80 000 Ar. Discount B: $-20\,\%$ then $-10\,\%$. Which is the better deal?
5. Thread · Method We know that $AB^2 + AC^2 = BC^2$. What conclusion can be drawn? Which property is used?

Sheet 51 Tue 16 Mar. 2027 *Marginal frequencies*

1. Warm-up · Quick Express $\frac{25}{40}$ as a percentage.
2. Core · Method Using the table from sheet 50: calculate the marginal frequency of girls and that of canteen users.
3. Core + · Quick The marginal frequency of girls is $55\,\%$. What is that of boys?
4. Review · Quick Calculate $(2a - b)^2$ for $a = 3$ and $b = -2$.
5. Thread · Quick $A(-1;2)$ and $B(3;2)$. What can be said about the line (AB)? Give AB.

Sheet 52 Thu 18 Mar. 2027 *Conditional frequencies*

1. Warm-up · Quick Give $\frac{15}{22}$ as a percentage rounded to the nearest tenth.
2. Core · Method With the same table: among the girls, what is the frequency of canteen users? Among the boys?
3. Core + · Distinguish Distinguish between "the frequency of girls among canteen users" and "the frequency of canteen users among girls".
4. Review · Construction Draw $[AB]$ of length 6 cm, then the circles with centres A and B and radius 4 cm, which meet at C and D. What type of quadrilateral is $ACBD$?
5. Thread · Quick Is 91 prime?

Sheet 53 Mon 22 Mar. 2027 *Completing a two-way table*

1. Warm-up · Quick Calculate 35 % of 80.
2. Core · Method 200 customers: 60 % are women; 25 % of the women and 50 % of the men buy online. Complete the two-way table.
3. Core + · Check A table states "row total: $45 + 55 = 100$" but "column total: $60 + 30 = 90$". What can be concluded?
4. Review · True/false True or false? Justify. After three consecutive "heads", the probability of getting "tails" on the next toss is greater than $\frac{1}{2}$.
5. Thread · Quick ABC is right-angled at A, $BC = 10$ cm, $\widehat{ABC} = 30°$. Calculate AC.

Sheet 54 Tue 23 Mar. 2027 *Real data: processing a file and filtering (AND, OR, NOT)*

1. Warm-up · Quick In a file of 1 000 rows, 400 satisfy A, 300 satisfy B and 100 satisfy both A and B. How many satisfy "A AND B"? "A OR B"?
2. Core · Method With the same data: how many satisfy "NOT A"? "A AND NOT B"?
3. Core + · Reading code What is the final value of `compte`?

```
compte = 0
for age in [15, 16, 17, 16, 18]:
    if age >= 16 and age < 18:
        compte = compte + 1
```

4. Review · Quick Give the decimal form of $\frac{7}{8}$, then the fraction in lowest terms equal to 0.12.
5. Thread · Formula → table $f(x) = x^2 - 2x$. Complete a table of values for $x \in \{0\,;\,1\,;\,2\,;\,3\}$.

Sheet 55 Thu 25 Mar. 2027 *Critical reading of statements from the media*

1. Warm-up · Quick A 10 % increase on 40: what is the final value?
2. Core · Critical thinking An article claims: "60 % of accidents involve sober drivers". Can we conclude that driving sober is dangerous?
3. Core + · True/false True or false? Justify. "Two smokers out of three are men" means that two men out of three smoke.
4. Review · True/false True or false? Justify. The median of a data set is always equal to its mean.
5. Thread · Reading code What is the value of `s` at the end?

```
s = 0
for i in range(1, 5):
    s = s + i
```